Hiding in the trees: Unmasking a cryptic genus of arboreal New Guinea frogs with phylogenetic taxonomy

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Abstract

Although cryptic species are known to underestimate biodiversity, cryptic genera may present greater challenges for evolutionary studies. Surprisingly, there remain long-standing groups that contain unrecognized polyphyly due to extreme similarity in morphology and lifestyle. A good example is the large frog genus Oreophryne whose monophyly was never questioned in its 130 year history. Recently, our group has shown it to be composed of two non-sister monophyletic clades using dense taxonomic sampling across the microhylid subfamily Asterophryinae. Here we show that the two arboreal clades are morphologically undiagnosible and define them using phylogenetic taxonomy. We include the type species O. senckenbergiana (= O. moluccensis), resolving which clade retains the name Oreophryne, and name the strikingly similar genus **Auparoparo gen. nov.** from the indigenous Papuan language Motu. Their geographic distributions implicate plate tectonic activity in diversification, with Oreophryne distributed from the Philippines to Indonesia, along central to south New Guinea and its satellite islands, whereas Auparoparo occurs in Sulawesi and southern New Guinea and satellite islands where there is range overlap. As evidenced by this case, groups with unrecognized polyphyly are still in use today, so we urge careful confirmation of monophyly for a nomenclature based on evolution that will promote sound evolutionary research.

Additional Keywords: cryptic taxa; homoplasy; indo-pacific; morphology; new classification; new genera; phylogenetic nomenclature; phylogenetics

Introduction

Oreophryne (Boettger, 1895) is the oldest and arguably the best studied genus of the hyperdiverse frog subfamily Asterophryinae (family Microhylidae). Although taxonomic instability is well known in the history of the subfamily Asterophryinae (reviewed in Rivera et al., 2017; Hill et al., 2022), Oreophryne seemed to be the exception. Well known for its arboreal morphology and lifestyle, the taxonomy of the genus Oreophryne as currently recognized has been stable over the past century. Recently it was discovered that the single large clade is, in fact, two independently evolved clades (reviewed in Hill et al., 2022). A natural question is how did this happen and why did it remain undiscovered for so long? One obvious answer is that their morphology is remarkably similar (Figure 1). Oreophryne are conspicuously arboreal and account for nearly all arboreal species of Asterophryinae (Günther, Richards, & Iskandar, 2001; Menzies, 2006), inhabiting lowland to montane rain forest, although there is a small group of diminutive, high-elevation, grassland species, a few of which are terrestrial (Zweifel, Cogger, & Richards, 2005; Günther & Richards, 2012; Putri et al., 2023). In short, Oreophryne are morphologically distinctive, consistent with their arboreal lifestyle, and appear in the field to be a natural group preserved by niche conservatism.

Oreophryne was united by a useful diagnostic character distinct from all other genera. Based on a pectoral anatomy with highly reduced and distinctly curved clavicles (Parker, 1934; Zweifel, 2003), hypothetical new species could be assigned to Oreophryne, and for modern studies, confirmed by phylogenetic analysis with other members of Oreophryne. Over the past 130 years the genus Oreophryne has grown to 75 currently recognized species, fully one-fifth of the total diversity of subfamily Asterophryinae and the largest of the ~16 genera (Hill et al., 2022; Frost, 2024). Oreophryne as currently recognized has the largest geographic distribution in the subfamily (Hill et al., 2023a), with species centered on mainland New Guinea and its offshore islands (Werner, 1898; Zweifel, 1956; Zweifel, Menzies, & Price, 2003; Kraus & Allison, 2004; Kraus & Allison, 2009; Kraus, 2013; Frost, 2024), but ranging westward to distant oceanic islands (Müller, 1894; Boulenger, 1896; Ahl, 1933; Putri et al., 2023) and northward to Indonesia (Boettger, 1895; Setiadi et al., 2010; Günther, Iskandar, & Richards, 2023) and the Southern Philippines Islands (Alcala, 1986; Balinton & de la Seña, 2015).

The apparently reliable diagnostic character, however, has never been demonstrated to be a synapomorphy, and recent studies have not found evidence for monophyly (Zweifel et al., 2005; Rivera et al., 2017). Recently, Hill et. al (2022, 2023b) established that Oreophryne sensu lato represents two reciprocally monophyletic groups that are not sister taxa (Figure 2), temporarily labeled "Oreophryne A" and "Oreophryne B". Furthermore, it is unknown whether it is possible to identify these groups with any morphological characters – there may in fact be no reliable diagnostic characters, rendering phylogenetic taxonomic definitions for each clade an imperative for evolutionary study. The aims of this study are to assess the phylogenetic position of the newly available type species of Oreophryne to name and describe the new genus, and to develop phylogenetic definitions for both clades in the context of the subfamily-level Asterophrinae phylogeny including representatives from all known genera. Furthermore, we analyze the morphological characteristics to assess whether morphological diagnosis is possible, and consider whether there are any additional features to assist collectors when identifying putative Oreophryne in the field (e.g., call type, geography, other characters), and discuss the implications of having misleading diagnostic characters for evolutionary studies. Inclusion of samples in molecular phylogenetics is not enough, and indeed it is necessary to include explicit tests of monophyly when proposing a new taxonomy.

Taxonomic History:

Boettger (1895) described the genus Oreoprhyne and the species Oreophryne senckenbergiana on the basis of three specimens from Halmahera Island, in the Maluku Archipelago west of New Guinea, that lacked procoracoids and had cartilaginous sterna. He also noted the presence of expanded finger and toe pads and transverse folds of skin in the palate. Parker (1934) noted that procoracoids were present, but were reduced and re-diagnosed the genus on the basis of this feature together with the presence of reduced clavicles, no omosternum, and a large cartilaginous sternum; all characters of the pectoral girdle and sternum. Zweifel (2003) noted that "the clavicles are tiny, slightly curved bones lying on (ventral to) the procoracoid cartilage and apart from the midline and the scapula." On the basis of these features there are now 75 recognized species in the genus (Frost, 2024).

We note that O. senckenbergiana was synonymized (Parker, 1934) with Microhyla achatina var. moluccensis described by Peters & Doria (1878) from Bacan and Ternate islands, based on morphology and the taxon is currently known as O. moluccensis. While Parker s (1934) methodology was standard at the time, genetic information has revealed that cryptic species are rampant in this group, with species readily separated by distance and oceanic barriers

(Hill et al., 2022, 2023b). Because O. senckenbergiana was described from Halmahera island, and Bacan and Ternate are smaller offshore islands of Halmahera, it is likely that they are different species that should be confirmed with molecular evidence.

Large scale molecular phylogenetic analysis with many ingroup samples was required to reveal that this morphologically united group contains two reciprocally monophyletic clades that are not sister taxa (Hill et al., 2022, 2023b). Early molecular phylogenetic studies suggested monophyly but only included a few species within a larger-scale study (Köhler & Günther, 2008 included 9 Oreophryne species; De Sa et al., 2012 included 4). Rivera et al. (2017) provided the first molecular evidence of a polyphyletic Oreophryne with 36 Oreophryne species within a larger phylogeny Asterophryinae which included 155 samples, followed by Tu et al. (2018)'s sparse supermatrix approach with 18 species of Oreophryne falling out in various places within a much larger phylogeny, but both of these lacked strong nodal support connecting the hypothesized clades to the backbone of the phylogeny and were thus inconclusive. The polyphyletic nature of Oreophryne sensu lato was not clearly demonstrated until Hill et. al (2022, 2023b) included 44 species of Oreophryne (with data 96% complete for sampled loci) within a 218 taxon phylogeny of Asterophryinae. This phylogeny is the most complete and robust phylogeny to date. The two clades, labeled "Oreophryne A" and "Oreophryne B" have deep histories of ~ 18MY and ~16MY, respectively, with "Oreophryne A" sister to the rest of the subfamily (Hill et al., 2022, 2023b).

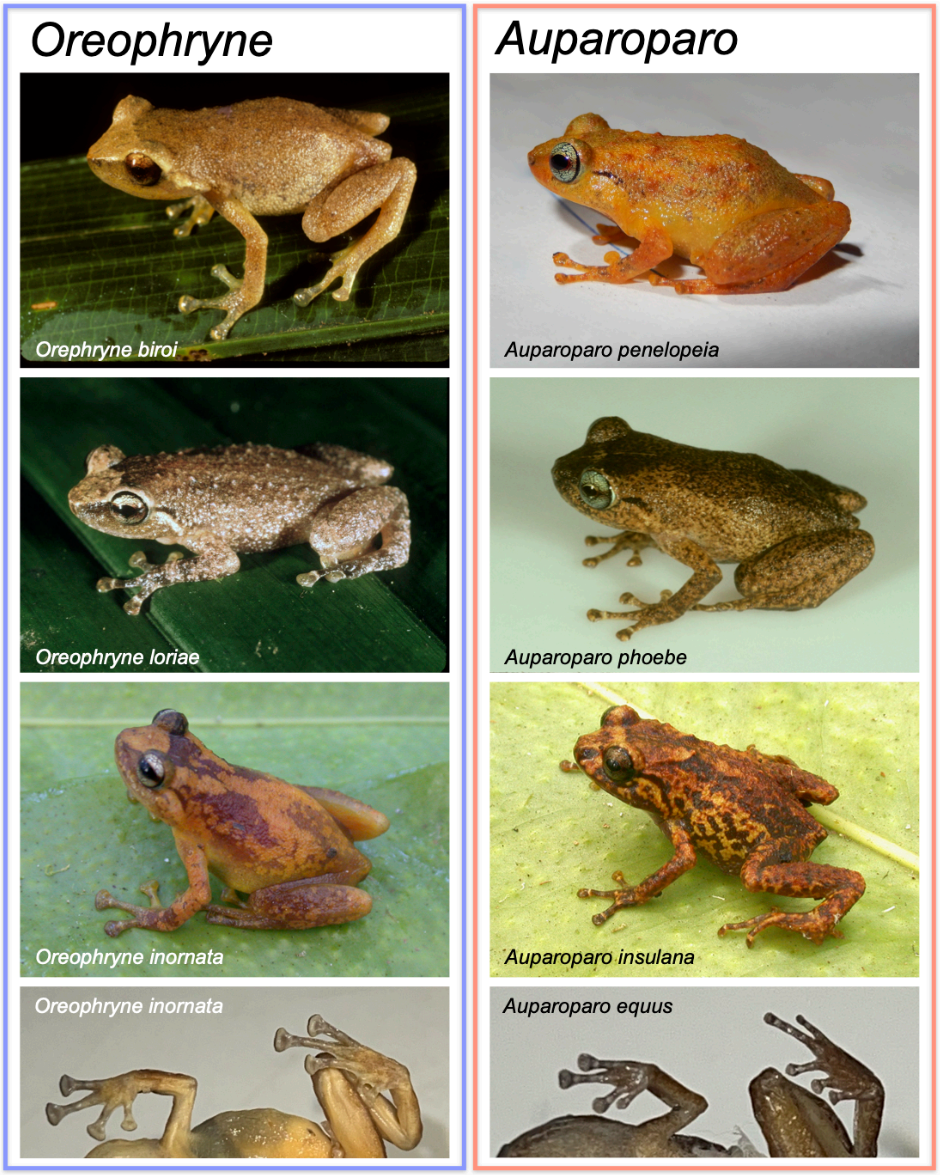

Figure 1: Photos of Oreophryne species on the left and Auparoparo species on the right. Both genera are aboreal with a few exceptions as noted in the text, and contain nearly all arboreal species within Asterophryinae. Comparison of finger and toe pad sizes in Oreophryne inornata BPBM16217 vs. Auparoparo equus BPBM15775, bottom row.

Materials and Methods

Taxon Sampling

Asterophryinae is an adaptive radiation with short branch lengths along the backbone separating currently recognized genera (Rivera et al., 2017; Hill et al., 2022, 2023b); we reconstruct the phylogeny of Oreophryne sensu lato within the larger context of the subfamily. We include 30 named species and 18 unnamed species of Oreophryne within a larger dataset of 234 samples of 203 species of Asterophryinae collected across New Guinea, the Philippines, and Indonesia with three outgroup species to root the phylogeny (Dyscophus antongilii, Scaphiophryne marmorata, and Platypelis grandis). Importantly, we include the type species Oreophryne senckenbergiana (Boettger, 1895) in a molecular phylogeny for the first time. The samples were collected in Halmahera, North Maluku Province, Indonesia by Iqbal Setiadi , Table 1, between Supu and Dodinga, two type localities for O. senkenbergiana (Boettger, 1895), which was later synonymized with O. moluccensis (Parker, 1934).

Table 1: Locality information for samples sequenced in this study. See E
1 for additional metadata.

ID	Species	Locality	LatLon	SIA
BJE01120	*Oreophryne moluccensis*	Kabupaten Halmahera Utara, Halmahera Island, North Maluku	1.8228, 127.8300	PV703775

ID	Species	Locality	LatLon	SIA
BJE01143	*Oreophryne moluccensis*	Kabupaten Halmahera Utara, Halmahera Island, North Maluku	1.7899, 127.8910	PV703776
BJE01144	*Oreophryne moluccensis*	Kabupaten Halmahera Utara, Halmahera Island, North Maluku	1.7899, 127.8910	PV703777

Molecular Sequences

We obtained complete DNA sequence data for five loci (three nuclear: Seventh in Absentia [SIA], Brain Derived Neurotrophic Factor [BDNF], Sodium Calcium Exchange subunit-1 [NXC-1], and two mitochondrial loci (Cytochrome oxidase b [CYTB], and NADH dehydrogenase subunit 4 [ND4] from three samples of the type species (not previously sequenced) and combined these with previously published data (Rivera et al., 2017; Hill et al., 2022, 2023b) for a dataset of 240 samples with 207 species and 2475 base pairs that is 96% complete (loci complete for the added samples). Sequence quality was evaluated using chromatogram quality scores, checks for stop codons indicating pseudogenes, BLAST searches (Ye et al., 2012), and concordance of pairwise similarities with other Asterophryinae sequences. Primer sequences and detailed methods including phylogenetic reconstruction are provided in (Hill et al., 2023b). We note that we excluded from the previous alignment (Hill et al., 2022, 2023b) two samples of Aphantophryne pansa (BPBM 5299 and BPBM 8312) which were deemed unreliable as they came from formalin-fixed material, and three samples which fell out in odd places that should be confirmed with new material prior to recommending taxonomic revision (BPBM 22708 Copiula tyleri, BPBM 38939 Copiula sp. 7, BPBM 37753 Paedophryne dekot). We note that genetic material from 47 species of Oreophryne sensu lato was not available for this molecular analysis.

Morphology, Behavior, and Geographic Distribution

We examined new morphological characters and reexamined traits compiled from the literature [Boettger (1895); Parker (1934); Kraus (2016); character states defined in Table 2] for all named species of Oreophryne in our phylogenetic dataset. Our sample included all species available in the collections of the Bishop Museum, Honolulu, Hawaii (BPBM): thirteen species of "Oreophryne A", eight species "Oreophryne B" (Table 3). Finger and toe pads were photographed and length and width of the toe tips were measured using ImageJ software. Trait information for the remaining species were gathered from the literature, six species of "Oreophryne A" and one of "Oreophryne B". In addition, we compiled information on call type from the literature and collection location from the respective collectors or the BPBM database (global positioning coordinates [GPS]; Table 3). We examined the ability of morphological and behavioral characters to provide a diagnosis to the two Oreophryne clades consistent with molecular phylogenetics.

Table 2: Description of traits surveyed.

Column*	Trait	Definition
1	Procoracoids	Size and shape of the procoracoid cartilages, CR= curved, reduced
2	Clavicles	Size and shape of the clavicles, CR= curved, reduced
3	Omasternum	Presence (+) or absence (-) of the omasternum
4	Sternum	Cartilaginous sternum or sternal plate
5	Call type	Qualitative description of the call type
6	Finger:toe width	The ratio of finger pad width to toe pad width
7	Pectoral ligament	Cartilaginous or ligamentous connection between scapula and procoracoid
8	Rel toe length	Relative length of 5th toe compared to length of 3rd toe (5<3, 5=3, 5>3)
9	Toe webbing	Toe webbing assessed on 5th toe (absent (-), basally webbed, well webbed)
10	Tympanum:eye diameter	Tympanum diameter relative to eye diameter (¼, □ , or ½)
	Pupil horizontal	Whether the pupil is horizontal
	Teeth	Whether teeth are present

Column*	Trait	Definition
	Palatal folds	Whether two horizontal palatal folds in front of the pharynx are present

*Corresponding column in Table 3. The last three traits were scored and are invariable across species, thus not tabluated in Table 3

Nomenclature

We conducted a comprehensive taxonomic literature review of diagnoses for all species known to belong to the clades "Oreophryne A" and "Oreophryne B" (Hill et al., 2022). We applied the ICZN code (International Commission on Zoological Nomenclature, 1999) as well as the Phylocode (De Queiroz, Kevin & Cantino, 2020). We support the nascent effort to assign indigenous names in taxonomy whenever possible (e.g., Gillman & Wright, 2020). One of us (B. Iova) proposed the novel genus name, Auparoparo, meaning "tree frog" in Motu, an indigenous language of Papua New Guinea, for "Oreophryne B".

Phylogenetic Reconstruction

We used Partitionfinder to find the best-fit evolutionary model followed by Maximum Likelihood (ML) phylogenetic reconstruction using IQTREE2 (Lanfear et al., 2012; Minh et al., 2020). Following Hill et al. (2022), we tested models partitioned by both locus and codon versus by loci only or by codon only (15 vs. 5 vs. 3 partition scheme) with merging of partitions allowed. We found following merging that a 9-partition scheme involving locus and codon models with merged partitions was superior by the Akaike Information Criterion (models by partition included in Supplementary Materials Table 1). Nodal support values for the ML tree were inferred using 1000 bootstrap replicates.

Phylogenetic Definition

We use a maximum crown-clade definition under the PhyloCode (De Queiroz & Gauthier, 1990, 1992; De Queiroz, Kevin & Cantino, 2020), whereby monophyletic clades are defined by the largest crown clade containing their reference species (in the case of genus-level clades, the type species for the genus or a proxy when molecular phylogenetic data is not available for the type species). Specifically, the largest crown clade originating with the most recent common ancestor of the type species for the genus-level clade, and all extant species that share a more recent common ancestor with that type species then they do with reference species of other clades.

Analyses, Visualization, and Data Availability

All analyses were conducted and graphics produced in the R Statistical Computing Environment (R Core Team, 2024) unless noted. Initial DNA sequence alignment was conducted using `MUSCLE` (Edgar, 2004) with manual adjustments in `Mesquite` (Maddison & Maddison, 2023). Phylogenetic reconstruction was conducted using the `IQTREE2` suite of software (Lanfear et al., 2012; Minh et al., 2020).

The known ranges of Oreophryne sensu lato were downloaded as shape files from the IUCN Red List database (IUCN, 2025), and plotted. Sites from collection locations for Oreophryne and Auparoparo were mapped using the `sf` and `ggplot2` packages (Wickham, 2016; Pebesma, 2018), overlaid on a basemap created using `rnaturalearth`, `grid`, and `ggspatial` packages (Dunnington, 2023; Massicotte & South, 2023). Tree graphics were produced with the `ggtree` package (Yu et al., 2017; Yu, 2020) supplemented by `ggplot2` (Wickham, 2016).

Results

Phylogeny and Phylogenetic Taxonomy

For our subfamily-level phylogeny, the best evolutionary model was a 9 partition scheme including a combination of locus by codon partitions (see Supplementary Table 1 for list partitions and models). The best set of partitions and models had an improvement of fit by 223 AIC units over the 15-partition (locus by codon) model, 3750 AIC units better than the 5 partition locus-only model, and 4230 AIC units improvement over the 3 partition codon-only model. All phylogenetic reconstructions reported here used the best-fit model (9-partition scheme) identified by IQTREE2.

We recovered a maximum likelihood topology that confirms that Oreophryne sensu lato is polyphyletic, with species falling into two independently-evolved, reciprocally monophyletic, highly-supported clades that are not sister groups Figure 2. Nearly all nodes are moderately or highly supported, with most of the weakly supported nodes falling along the backbone where there are very short branch lengths as characteristic of an adaptive radiation.

Furthermore, these findings demonstrate that the crown clade descended from node A should retain the name Oreophryne, as the type species, O. senckenbergiana, from North Maluku Island, Indonesia, is situated within Oreophryne with high support (Figure 2). We propose the new clade name Auparoparo for the crown clade descended from node B. Oreophryne is the earliest diverging genus-level clade in the Papuan subfamily, with genera Cophixalus and Paedophryne arising before Auparoparo, which split from the remaining genera of Asterophryinae.

Morphology, Behavior, and Ranked Taxonomy

We find no morphological synapomorphies that distinguish Auparoparo from Oreophryne. Oreophryne sensu lato is diagnosed by characters of the pectoral girdle: posessing highly reduced, curved procoracoid cartilages and clavicles, and lacking an omosternum (Parker, 1934; Zweifel, 2003). We found that this suite of characters is also present in Auparoparo, and therefore is not diagnostic for either Oreophryne or Auparoparo (Table 3).

Two traits imperfectly segregate between the clades, relative finger and toe pad widths and call type (Table 3), which may assist in tentative identification. Both Oreophryne and Auparoparo are arboreal clades with expanded digital pads. Species of Oreophryne have finger and toe pads generally of roughly equal width and most species of Auparoparo have visibly larger finger pads, with finger pads roughly 20-30% wider than toes. Both Oreophryne and Auparoparo use a variety of call types, with “honks” or “peeps” used only by Oreophryne, whereas “whinny” calls are used only by Auparoparo. A minority of species in both groups use the “rattle” call type. However, both toe pad width and call type are variable traits with some overlap and thus not unambiguosly diagnostic. Furthermore, possesssion of expanded digital pads is clearly an adaptive trait for arboreality and prone to homoplasy.

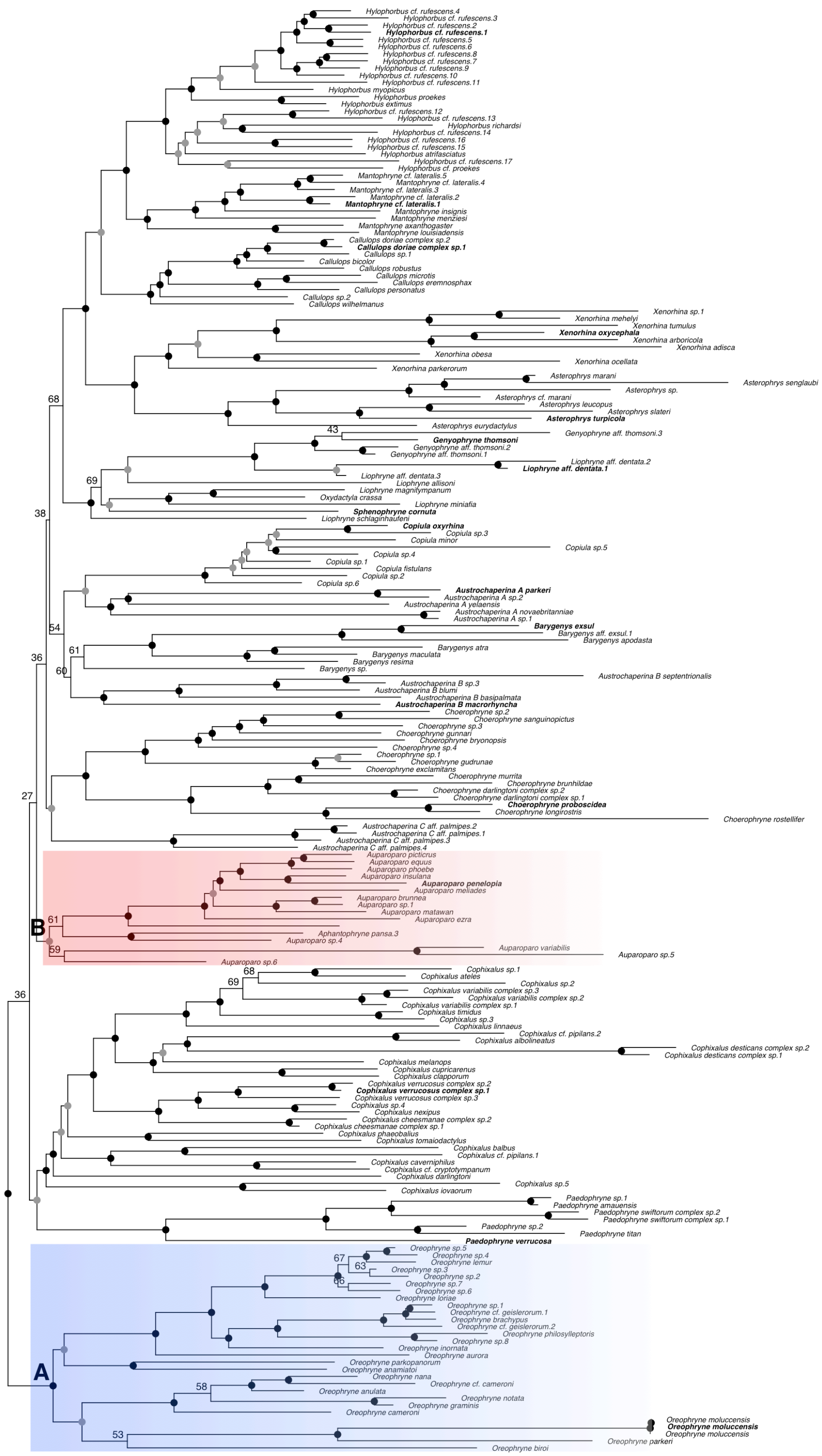

Figure 2: Molecular phyogenetic hypothesis of Asterophryinae showing polyphyly of Oreophryne sensu lato, into two reciprocally monophyletic clades that are not sister taxa. Node A is Oreophryne (blue clade) and node B is Auparoparo (red clade) based on the position of the type species Oreophryne senkenbergiana=O. moluccensis. Nodal support: black dots indicate ML boostrap support >= 95%, grey dots >= 75%, numeric values given for nodes below 75% support. Outgroup taxa excluded. Taxa in bold are the type species for each genus (or a proxy species used as an internal reference).

Table 3: Morphological character m

		(1)	(2)	(3)	(4)	(5)	(6)	(7)
Genus	**species**	**proc**	**clav**	**omo**	**ster**	**call type**	**fing: toe width**	**pec lig**
Oreophryne	*anamiatoi*	CR	CR	–	cart	chuckle or cackle	1.2	cart
	anulata[1]	CR	CR	–	cart		slightly larger	
	aurora	CR	CR	–	cart	harsh croaks		lig
	biroi	CR	CR	–	cart	rattle	1.1	lig
	brachypus	CR	CR	–	cart	long squeak	1.1	lig
	cameroni	CR	CR	–	cart	peep		cart
	frontifasciata[1]	CR	CR	–	cart		slightly larger	cart
	geislerorum	CR	CR	–	cart	rattle		lig
	graminis[1]	CR	CR	–	cart	peep	approx equal	cart
	inornata	CR	CR	–	cart	honk	1.0	lig
	lemur	CR	CR	–	cart	honk		lig

		(1)	(2)	(3)	(4)	(5)	(6)	(7)
Genus	**species**	**proc**	**clav**	**omo**	**ster**	**call type**	**fing: toe width**	**pec lig**
	loriae	CR	CR	–	cart	honk		lig
	moluccensis[1]	CR	CR	–	cart		slightly larger	
	nana[1]	CR	CR	–	cart		approx equal	cart
	notata	CR	CR	–	cart	peep		cart
	parkeri	CR	CR	–	cart	peep		lig
	parkopanorum	CR	CR	–	cart			cart
	philosylleptoris	CR	CR	–	cart	honk	1.0	lig
Auparoparo[2]	*brunnea*[1]	CR	CR	–	cart	rattle	larger	lig
	equus	CR	CR	–	cart	whinny	1.3	lig
	ezra	CR	CR	–	cart	chatter		lig
	insulana	CR	CR	–	cart	whinny		lig
	matawan	CR	CR	–	cart	rattle		lig

		(1)	(2)	(3)	(4)	(5)	(6)	(7)
Genus	species	proc	clav	omo	ster	call type	fing: toe width	pec lig
	meliades	CR	CR	–	cart	whinny	1.2	lig
	penelopeia	CR	CR	–	cart	whinny	1.3	lig
	phoebe	CR	CR	–	cart	whinny	1.2	cart
	picticrus	CR	CR	–	cart	whinny	1.3	lig
	variabilis[1]	CR	CR	–	cart		much larger	cart

[1]Character states obtained from the literature, specimens not available for examination in this study.

[2]Hill et al. 2022 places *Aphantophryne pansa* within *Auparoparo*, but the status of *Aphantophryne* requires further study with samp

*External characters were examined from specimens, internal characters and pupil shape taken from the literature.

Geographic Distribution

Oreophryne has a wide distribution with species spanning from the southern Philippines to the Papuan Louisade Archipelago (Figure 3, localities for Oreophryne species included in our phylogeny are mapped with blue triangles). This range encompasses the additional species from the type locality in Indonesia. Auparoparo is mainly clustered in the southeastern tip of mainland Papua New Guinea and its satellite islands with two species in Sulawesi, an Indonesian island (Figure 3, Auparoparo localities mapped in red circles). We also mapped Oreophryne sensu lato ranges obtained from the IUCN Red List (IUCN, 2025), which require phylogenetic diagnosis. Note the Indonesian region of New Guinea is difficult to attain samples from and it is unknown whether these groups inhabit this connective area.

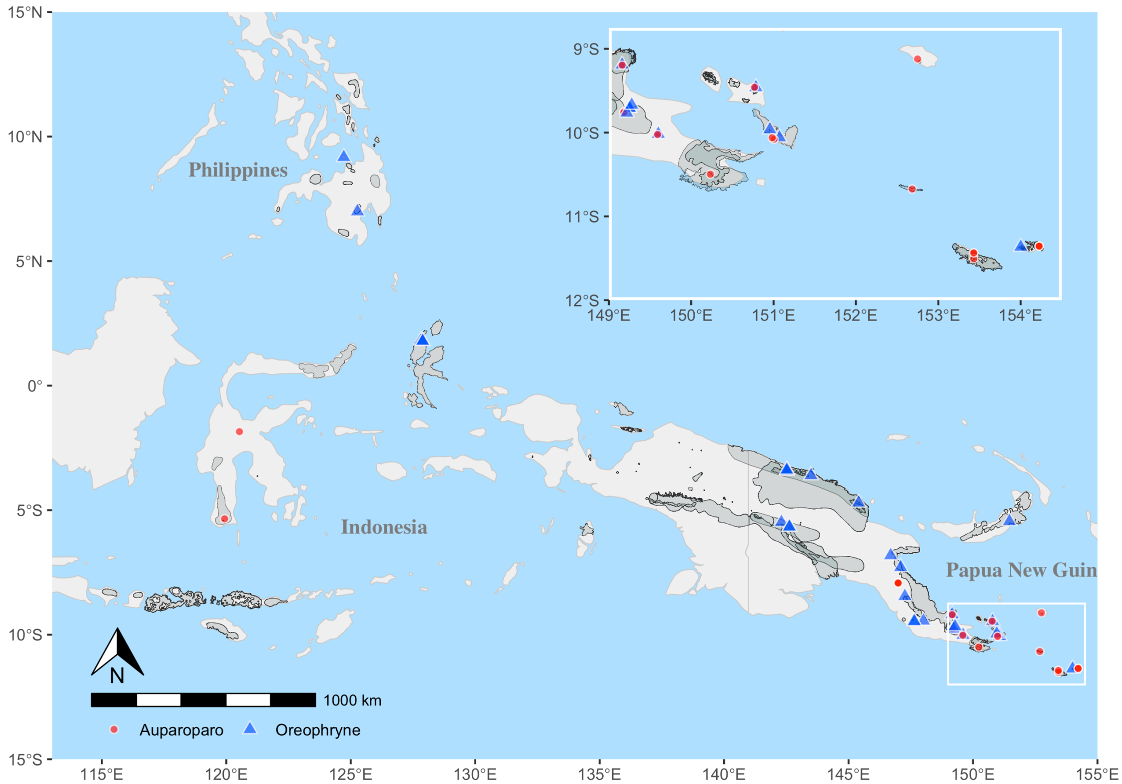

Figure 3: The known distribution of Oreophryne as currently recognized is plotted in grey polygons (data source IUCN, 2025), with points indicating Auparoparo (red) and Oreophryne (blue) as assigned by our phylogenetic analysis Figure 2. Note that there is range overlap along the East Papuan Composite Terrane and on some offshore islands (see inset).

Taxonomy

We use maximum crown-clade definitions for designating Oreophryne and Auparoparo because the monophyly of these taxa are strongly supported, and the relationships of these taxa to each other and to other genera of Asterophryinae are well-supported (this study, Hill et al., 2022, 2023b), as has been done for other anuran taxa (Brown et al., 2015) following the methods of phylogenetic nomenclature (De Queiroz & Gauthier, 1990, 1992, 1994). That is, we adopt phylogenetic definitions of these genera that explicitly identify them as clades. Specifically, we define each genus to include all descendants of the most recent common ancestor of the type species and all other extant species that are more closely related to it than to the type species of the other Asterophryine genera recognized here. We retain the name Oreophryne for species included in the phylogenetic redefinition of Oreophryne, and new clade name Auparoparo for newly designated species of Auparoparo. The previously defined polyphyletic Oreophryne is indicated by "sensu lato", which contains many species which require molecular phylogenetic assignment.

Family Microhylidae Günther 1858

Subfamily Asterophryinae Günther 1858

Genus Oreophryne Boettger 1895 amended

Microhyla in part (Peters 1878)

Callula in part (Horst 1883)

Oreophryne Boettger 1895, by monotypy, Type species: Oreophryne senckenbergiana Boettger 1895. (= Microhyla achatina var. moluccensis Peters 1878) Placed on Official List of Generic

Names in Zoology by Opinion 1266, Anonymous, 1984.

Sphenophryne in part (Méhely 1897)

Phrynixalus in part (Stejneger 1908)

Mehelyia Wandolleck 1911, Type species: Meheylia lineata Wandolleck 1911 (= Sphenophryne biroi Mehely 1897), by subsequent designation of Parker 1934. Synonymy by Nieden (1926).

Cophixalus in part (Boettger 1892)

Chaperina in part (Taylor 1920)

Diagnosis: We note that there are no synapomorphies in adult Oreophryne that differ from the morphologically similar genus Auparoparo. Diagnoses should be made by molecular phylogenetic analysis, based on the phylogenetic definition below. Oreophryne differs from most other genera in subfamily Asterophryinae, but not from Auparoparo, by the combination of (1) presence of highly reduced clavicles (vs absence or elongate) which have a distinctly curved shape; (2) presence of well developed digital pads, often with finger and toe pads of similar size; (3) arboreal lifestyle (perch height >2m above ground); (4) most species range in body size from 16mm (O. graminis, Günther & Richards, 2012) to 30mm, with a few larger species ~40mm (O. inornata); (5) small to invisible tympanum; (6) large eyes with horizontal pupils; (7) occasional presence of toe webbing; (8) thin, elongated limbs; and (9) in many species males that call with "honk" or "peep" types or in some species "rattle" type. Some species of Oreophryne may be distinguished from Auparoparo by roughly equal finger to toe pad ratio and males calling with "honk" or "peep" but not "whinny" call types.

Phylogenetic definition: Oreophryne Boettger 1895 (Converted Clade Name) is defined as the largest crown clade including Oreophryne senckenbergiana Boettger 1895 and excluding the type species of the other genus-ranked clades listed in Table Table 4.

Table 4: Genus-level clades of Asterophryinae and their internal specifiers (type species or proxies).

Genus-level clade	Internal Specifier (Type Species or proxy)
Asterophrys	*Asterophrys turpicola* (Schlegel 1837)
Auparoparo	*Auparoparo penelopeia* (Kraus 2016)
*Austrochaperina**	*Austrochaperina palmipes* (Zweifel 1956)
*Austrochaperina**	*Austrochaperina robusta* Fry 1912
Barygenys	*Barygenys cheesmanae* Parker 1936
Callulops	*Callulops doriae* Boulenger 1888
Choerophryne	*Choerophryne proboscidea* Van Kampen 1914
Cophixalus	*Cophixalus verrucosus* (Boulenger, 1898)
Copiula	*Copiula oxyrhina* (Boulenger 1898)
*Genyophryne**	*Genyophryne thomsoni* Boulenger, 1890

Genus-level clade	Internal Specifier (Type Species or proxy)
Hylophorbus	*Hylophorbus rufescens* Macleay 1878
Mantophyrne	*Mantophryne lateralis* Boulenger 1897
*Liophryne**	*Liophryne rhododactyla* (Boulenger 1897)
Oreophryne	*Oreophryne senckenbergiana* Boettger 1895 (= *O. moluccensis* (Peters and Doria 1878))
Oxydactyla	*Oxydactyla brevicrus* (Van Kampen 1913)
Paedophryne	*Paedophryne kathismaphlox* Kraus 2010
*Sphenophryne**	*Sphenophryne cornuta* Peters and Doria 1878
Xenorhina	*Xenorhina oxycephala* (Schlegel 1858)

*We note that the monophyly of *Austrochaperina, Liophryne, Oxydactyla*, and *Sphenophryne* (and reciprocal monophyly of *Genyop*

[*Genyophryne thomsoni* Boulenger, 1890] with respect to *Liophryne + Oxytactyla + Sphenophryne*) has not yet been established.

Austrochaperina is polyphyletic in recent phylogenies with possibly three monophyletic clades that are not sister taxa (Figure 2).

Composition: Our results support the retention of 17 species in Oreophryne with molecular evidence (Table 5): Oreophryne anamiatoi Kraus 2009, O. anulata (Stejneger 1908), O. aurora Kraus 2016, O. biroi Mehely 1897, O. brachypus Werner 1898, O. cameroni Kraus

2013, O. geislerorum Boettger 1892, O. graminis Gunther 2012, O. inornata Zweifel 1956, O. lemur Kraus 2016, O. loriae Boulenger 1898, O. nana Brown 1967, O. notata Zweifel 2003, O. parkeri Loveridge 1955, O. parkopanorum Kraus 2013, O. philosylleptoris Kraus 2016, and O. moluccensis (Peters 1878). A number of species were not available for our study but should molecular evidence demonstrate their relationship within Oreophryne, those species should also retain their generic identity.

Distribution: The range of confirmed Oreophryne species in this dataset remains nearly as broad as previously defined (Figure 3), from the Philippines and Indonesia to the south-eastern most tip of mainland New Guinea and several near-shore islands (Fergusson, and Normanby Is.) and the distant Rossel Island of the Louisiade Archipelago. However, the transfer of species which occupy Woodlark Islands and Misima and Sudest Islands of the Louisaide Archipelago reduces the known range of Oreophryne. Woodlark Island is inhabited by O. phoebe [=Auparoparo phoebe], Misima Island is inhabited by O. picticrus [=Auparoparo picticrus], and Sudest Island is inhabited by O. ezra and O. meliades [= Auparoparo ezra and Auparoparo meliades].

Genus Auparoparo gen. nov.

Oreophryne in part (Boettger, 1895)

Sphenophryne in part (Boulenger, 1897)

Type species Oreophryne penelopeia (Kraus, 2016) [=Auparoparo penelopeia]

Diagnosis: Currently, there are no known synapomorphies in adult Auparoparo that differ from the morphologically similar genus Oreophryne. Diagnoses should be made by molecular phylogenetic analysis, based on the phylogenetic definition below. Auparoparo differs from most other genera in subfamily Asterophryinae, but not from Oreophryne, by the combination of (1) presence of highly reduced clavicles (vs absence or elongate) which have a distinctly curved shape; (2) presence of well developed digital discs, with finger discs notably wider than toe discs (typically 20% or more); (3) arboreal lifestyle (perch height >2m above ground); (4) small to moderately large body size in known species, from 16mm (A. brunnea Kraus 2017) to 25 mm (A. picticrus Kraus 2016); (5) small to invisible tympanum; (6) large eyes with horizontal pupils; (7) occasional presence of toe webbing; (8) and thin, elongated limbs; and (9) and in many species males that call with a "whinny" call type or in some species "rattle" type. Some species of Auparoparo may be distinguished from Oreophryne by larger finger pads than toe pads and males calling with "whinny" but not "honk" or "peep" call types.

Phylogenetic definition: Auparoparo (New Clade Name) applies to the largest crown clade including Auparoparo penelopia (Kraus 2016) and excluding the type species of the other genus-ranked clades listed in Table 4.

Composition: We transfer eleven named species from Oreophryne into Auparoparo (Table 5): Auparoparo brunnea (Kraus, 2017), A. equus (Kraus, 2016), A. ezra (Kraus & Allison, 2009), A. insulana (Zweifel, 1956), A. matawan (Kraus, 2016), A. meliades (Kraus, 2016), A. penelopeia (Kraus, 2016), A. phoebe (Kraus, 2017), A. picticrus (Kraus, 2016), A. penelopia (Kraus, 2016), A. variabilis (Boulenger, 1896).

Distribution: Southeastern tip of mainland Papua New Guinea and satellite islands (Fergusson, Misima, Normanby, Rossel, Sudest, Woodlark Is.), Sulawesi.

Etymology: The generic name Auparoparo comes from the indigenous language, Motu, which is widely spoken in the region of Papua New Guinea in which this genus is found. It is a combination of the words "au" (tree) and "paro paro" (frog). Motu is not a gendered language, however, for nomenclatural purposes we designate the genus Auparoparo as feminine, thereby preserving the gender of existing species names.

Discussion

Our phylogenetic results (Hill et al., 2022; 2023b, this study) provide clarity on the classification of the problematic genus Oreophryne as currently recognized. We confirm that Oreophryne sensu lato is polyphyletic and consists of two reciprocally monophyletic clades that are not sister taxa (Figure 2). Here we designate "Oreophryne A" in Hill et al. (2022) as the genus Oreophryne because it contains the type species Oreophryne senckenbergiana Boettger 1895 with high support. Oreophryne is the earliest diverged genus in subfamily Asterophryinae originating approximately 18 MYA (Hill et al., 2022), and is sister to a clade composed of all the other genera. The second clade, "Oreophryne B" in Hill et al. (2022), is highly supported as a separate, non-sister genus, which originated approximately 16 MYA and contains no known species that are the types of any previously proposed genera. Thus, we designate this genus as Auparoparo, a tribute to the Motu language of the indigenous peoples of the region.

The Difficulty with Cryptic Genera: Can we tell them apart?

Perhaps the greatest testament to the crypticism of *Oreophryne* vs. *Auparoparo* is that no taxonomist was able to distinguish that they are separate clades, independently evolved, in the 130 years between their discovery and the present classification [Boettger (1895); this study]. These species are indeed morphologically very similar (Figure 1) and our results provide some clues as to why *Oreophryne* taxonomy has been so difficult. We found morphological synapomorphies for neither *Auparoparo* nor *Oreophryne*, and moreover, no morphological features that can diagnose either clade.

The traditional basis for assigning species to *Oreophryne* has primarily relied on a signature pectoral anatomy (whether original or revised diagnosis, Parker, 1934; Zweifel, 2003). Our phylogenetic results clearly show that this trait appears more than once, in both *Oreophryne* and *Auparaparo*, is either a symplesiomorphy (shared ancestral character) or a result of convergent or parallel evolution, and perhaps adaptive for arboreality. Thus, the pectoral anatomy is not a reliable diagnostic character in differentiating between *Oreophryne* and *Auparoparo*. In studies across a wide diversity of anurans, Soliz *et al.* (2025) found that anatomical variation of the pectoral girdle corresponds with functional and ecological lifestyle demands, whereas Engelkes *et al.* (2020) found strong phylogenetic influence in the shape of the pectoral girdle. The pectoral girdle would be excellent for further study within Asterophryinae, as there are high elevation terrestrial species of *Oreophryne* sensu lato that possess the same pectoral trait (Zweifel *et al.*, 2005), as well as a few arboreal species within other genera (*Cophixalus*, *Asterophrys*, *Xenorhina*) with unstudied pectoral morphology.

We further rexamined additional characters previously used to diagnose Oreophryne as well as characters conventionally used to diagnose species of frogs. We conclude that pectoral girdle connector type (Parker, 1934) and relative lengths of the 3rd and 5th toe are uninformative and the degree of toe webbing is variable. It was previously suggested that species in Oreophryne sensu lato could be divided into two groups based on possessing a cartilaginous or ligamentous connection between the procoracoid cartilages and the scapula (Kraus, 2013), with roughly half possessing each type (Parker, 1934). However, we found that cartilaginous and ligamentous connectors are common across both genera (Table 3); therefore, these traits are not synapmorphies for Oreophryne or Auparoparo and do not indicate monophyly for these clades. Of species with toe webbing, Oreophryne tends to be described as having "well webbed" toes compared to Auparoparo's "basal" webbing (Table 3). However, the presence of intraspecific variability combined with subjectivity in judging relative landmarks (e.g., "reaching the penultimate tubercle", "webbed 1/3 length of 5th toe", etc.), makes this feature difficult to standardize for generic diagnostic use. Many of the morphological traits proposed by Parker (1934) and Zweifel (2003) may in fact be a result of homoplasy via convergent lifestyle or parallel evolution, and while they may prove useful for ecomorphological identification or species diagnoses in combination with other traits, they are not informative for generic diagnosis.

We found only two characters with potential for imperfectly distinguishing the genera: relative digital pad size (Parker, 1934; Zweifel, 1956), and call type (e.g., Kraus, 2016). The relative widths of the digit pads differs between genera: Auparoparo tends to have visibly larger finger pads in comparison to toe pads, whereas in Oreophryne the finger and toe pads are roughly equal in size (Figure 1, bottom row), although there is some overlap in finger to toe pad ratio (Table 3). In some cases, Auparoparo's finger pads display T-shapes while the toes are often rounder and narrower. Whether there is any performance advantage in larger finger pads compared to toes is unknown. Furthermore, a handful of high-elevation Oreophryne species not included in our phylogeny and yet to be diagnosed by molecular phylogenetic analysis have evolved terrestrial or fossorial habits and possess smaller digital pads compared to their presumed arboreal relatives (Zweifel et al., 2005).

Call features are increasingly used for diagnosing species of Oreophryne, with advertisement call types commonly categorized as rattle, honk, whinny, peeping, or clicking (Kraus, 2016). We found honk or peep call types are only found in Oroephryne species in our sample, while whinny calls are only found in Auparoparo species, and rattle calls are found in both genera (Table 3). We note that several authors described species of both genera, thus the difference between genera cannot be attributed to the investigator. However, more complete sampling may add further complexity. As far as we are aware there has been no comprehensive acoustic analysis across Oreophryne sensu lato. More research is needed to explore call diversity both between species and across genera to understand call evolution and whether there any features that are distinctive of genera.

Geographic Distribution

Both Oreophryne and Auparoparo are geographically centered in the mainland New Guinea. Oreophryne extends southward to the Louisiade Archipelago and northward to the South

Philippine Islands (Hill et al., 2023b). Addition of O. moluccensis expands the known range eastward to North Maluku, Indonesia (Figure 3).

The known distribution of Auparoparo is mainly clustered in the southeastern tip of mainland Papua New Guinea (in the East Papuan Composite Terrane) and satellite islands with two species in Sulawesi, Indonesia. This suggests the possiblity of of interesting plate tectonics resulting in this biogeographic distribution. Exploration of intermediate localities may result in as yet undescribed or undiscovered species of Auparoparo. The geographic range is sure to grow with additional phylogenetic sampling. For example Oreophryne (sensu lato) chlorops (Günther et al., 2023) is a very large (SVL > 40mm) species recently discovered from mid-elevation mountains of the bird s neck region of Western Papua with clearly enlarged finger pads relative to toe pads. If confirmed as Auparoparo, it would significantly expand the range of Auparoparo across New Guinea Island.

We found geographic range overlap between Oreophryne and Auparoparo at multiple localities: on the mainland along the Mt. Dayman Foothills, at the Mt. Simpson Massif, Mt. Trafalgar, and on the oceanic islands of Normanby and Fergusson (both nearshore islands), and Rossel island (hundreds of kilometers offshore), which presents interesting questions regarding biogeography as well as their ecological relationships. For example, Oreophryne inornata and Auparoparo insulana have been collected at the same site (Goodenough Island) and even the same microhabitats (Zweifel, 1956). Do these species occupy the same niche, and what is the temporal history of each colonization? Are both genera also present in the nearby islands of Normanby and Fergusson (all part of the D'Entrecasteaux Archipelago)? Obtaining additional geographical samples for inclusion in phylogenetic analysis should be a high priority to gain a more complete understanding of the geographic distribution and evolutionary history of both Oreophryne and Auparoparo.

How to Avoid a Misleading Classification

We adopt an explicitly phylogenetic framework for the new taxonomy of the genera Oreophryne and Auparoparo, and to our knowledge, this is the first time that phylogenetic nomenclature has been applied within the subfamily Asterophryinae. A phylogenetic definition of genera is advantageous especially because Asterophryinae is an adaptive radiation. While we are increasingly confident that at least 16 genera are indeed monophyletic, nevertheless, very short branch lengths separate many of the genera along the backbone [Hill et al. (2022); this study]. A maximum-crown-clade definition with multiple external specifiers is robust to uncertainty (i.e., rearrangements of order) along the backbone. We define each genus as the largest crown clade including the internal specifier species for the clade (i.e., the type species of the genus) but not the internal specifier species of other asterophryine genera. For long-studied groups such as Asterophryinae, it may not be possible to obtain a genetic sample for type species that were collected in the 1800 s and early 1900 s, as these locations may be uncertain and/or remote, it may be necessary to use a genetic sample from another species of the genus as a proxy for the type species. Nevertheless, is increasingly clear that for Oreophryne and Auparoparo, a molecular phylogenetic approach is required for final confirmation of taxonomy.

Furthermore, a phylogenetic definition provides guidance for robust classification. It is imperative that when assiging new species to either Oreophryne or Auparoparo, molecular confirmation requires – at minimum – inclusion of members of the other clade to correctly identify the genus level clade. For example, a phylogenetic analysis with only Oreophryne sensu lato and outgroups to Asterophryinae will show incorrect grouping of Oreophryne and Auparoparo species because of the lack of intervening species assigned to other Asterophryinae genera. An additional difficulty here is that each clade, Oreophryne and Auparoparo, contains subclades with deep histories (Hill et al., 2023a), so we recommend the inclusion of additional intervening species assigned to other Asterophryinae genera, particularly Cophixalus and Paedophryne, to reduce the potential for long-branch attraction and more clearly resolve the "boundaries" of each monophyletic clade.

This study furthermore highlights the importance of taxonomic sampling for robust phylogenetic results. Developing an orderly taxonomy in a clade with over 350 named species (with a sizeable proportion of specimens in museum collections that are yet to be described, A. Allison, unpublished) would be that much more challenging without the benefit of generic diagnoses that follow evolutionary history.

Conclusion: Why the Names Matter

Clarifying the generic structure also facilitates appreciation for the patterns of diversification within Asterophryinae. The evolutionary history of this subfamily is characterized by several interesting trends across genera: 1) a burst of ecological and species diversification leading to generic-level clades occurring early in the history of the subfamily (Hill et al., 2022), 2) a tendency toward niche conservatism within genera, and 3) simultaneous dispersal of multiple genera to offshore islands (Hill et al., 2022, 2023a) revealing the very strong influence of plate tectonics for explaining the historical dispersal patterns of this group. Finally, without distinct generic names for the two clades represented by Oreophyne and Auparoparo, it becomes much more difficult to appreciate independently evolutionary histories embodied by each clade, that aboreality has evolved at least twice in two large clades of Asterophyrinae, and would make it very difficult to appreciate the impressive paralleism embodied by these speciose and successful genera. An accurate and informative taxonomy serves to advance our understanding of Papuan microhylids and their very interesting evolutionary history leading to further discovery.

Lastly, we emphasize that future naming of species, genera, and higher taxa should seek to recognize the language of the indigenous stewards of the land and ecosystem from which they were discovered (Gillman & Wright, 2020). This is especially true of Asterophryinae frogs, whose awe-inspiring diversity mirrors the rich cultural beauty and historical complexity of Papuan people.

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Allison R. Fisher (Conceptualization, Methodology, Data Collecting, Analysis, Writing, Reviewing, and Editing), **Ke Cao** (Data Collecting, Analysis, Writing, Editing), **Allen Allison** (Conceptualization, Writing, Reviewing, and Editing), **Bulisa Iova** (Collections, Land Access, Writing), and **Marguerite A. Butler** (Conceptualization, Analysis, Visualization, Supervision, Writing, Editing)

Supplementary Data

Table 5: Classification of Auparoparo and Oreophryne based on phy
two mitochondrial gene fragments (CytB, ND4) and three n
NXC). Incertae sedis taxa require phylogenetic evidence for

Original designation	Citation	Previous generic placement	Cu pla
Oreophryne anamiatoi	Kraus & Allison, 2009	*Oreophryne*	*Or*
Phrynixalus anulatus	Stejneger, 1908	*Oreophryne*	*Or*
Oreophryne aurora	Kraus, 2016	*Oreophryne*	*Or*
Sphenophryne biroi	Méhely, 1897	*Oreophryne*	*Or*
Hylella brachypus	Werner, 1898	*Oreophryne*	*Or*
Oreophryne cameroni	Kraus, 2013	*Oreophryne*	*Or*
Cophixalus geislerorum	Boettger, 1892	*Oreophryne*	*Or*
Oreophryne graminis	Günther & Richards, 2012	*Oreophryne*	*Or*
Oreophryne inornata	Zweifel, 1956	*Oreophryne*	*Or*
Oreophryne lemur	Kraus, 2016	*Oreophryne*	*Or*

Original designation	Citation	Previous generic placement	Cu pla
Sphenophryne loriae	Boulenger, 1898	*Oreophryne*	*Or*
Microhyla achatina	var. moluccensis Peters and Doria, 1878	*Oreophryne*	*Or*
Oreophryne nana	Brown & Alcala, 1967	*Oreophryne*	*Or*
Oreophryne notata	Zweifel, 2003	*Oreophryne*	*Or*
Oreophryne parkeri	Loveridge, 1955	*Oreophryne*	*Or*
Oreophryne parkopanorum	Kraus, 2013	*Oreophryne*	*Or*
Oreophryne philosylleptoris	Kraus, 2016	*Oreophryne*	*Or*
Oreophryne brunnea	Kraus, 2017	*Oreophryne*	*Au*
Oreophryne equus	Kraus, 2016	*Oreophryne*	*Au*
Oreophryne ezra	Kraus & Allison, 2009	*Oreophryne*	*Au*
Oreophryne gagneorum	Kraus, 2013	*Oreophryne*	*Au*
Oreophryne insulana	Zweifel, 1956	*Oreophryne*	*Au*

Original designation	Citation	Previous generic placement	Cu pla
Oreophryne matawan	Kraus, 2016	*Oreophryne*	*Au*
Oreophryne meliades	Kraus, 2016	*Oreophryne*	*Au*
Oreophryne penelopeia	Kraus, 2016	*Oreophryne*	*Au*
Oreophryne phoebe	Kraus, 2017	*Oreophryne*	*Au*
Oreophryne picticrus	Kraus, 2016	*Oreophryne*	*Au*
Sphenophryne variabilis	Boulenger, 1896	*Oreophryne*	*Au*
Oreophryne albitympanum	Günther, Richards, and Tjaturadi, 2018	*Oreophryne*	*inc*
Oreophryne albomaculata	Günther, Richards, and Dahl, 2014	*Oreophryne*	*inc*
Sphenophryne albopunctata	Van Kampen, 1909	*Oreophryne*	*inc*
Oreophryne alticola	Zweifel, Cogger, and Richards, 2005	*Oreophryne*	*inc*
Oreophryne ampelos	Kraus, 2011	*Oreophryne*	*inc*
Oreophryne anser	Kraus, 2016	*Oreophryne*	*inc*

Original designation	Citation	Previous generic placement	Cu pla
Sphenophryne anthonyi	Boulenger, 1897	*Oreophryne*	*inc*
Oreophryne asplenicola	Günther, 2003	*Oreophryne*	*inc*
Oreophryne atrigularis	Günther, Richards, and Iskandar, 2001	*Oreophryne*	*inc*
Oreophryne banshee	Kraus, 2016	*Oreophryne*	*inc*
Oreophryne brevicrus	Zweifel, 1956	*Oreophryne*	*inc*
Oreophryne brevirostris	Zweifel, Cogger, and Richards, 2005	*Oreophryne*	*inc*
Sphenophryne celebensis	Müller, 1894	*Oreophryne*	*inc*
Oreophryne choerophrynoides	Günther, 2015	*Oreophryne*	*inc*
Oreophryne chlorops	Günther, Iskandar, and Richards, 2023	*Oreophryne*	*inc*
Oreophryne clamata	Günther, 2003	*Oreophryne*	*inc*
Cophixalus crucifer	Van Kampen, 1913	*Oreophryne*	*inc*
Oreophryne curator	Günther, Richards, and Dahl, 2014	*Oreophryne*	*inc*

Original designation	Citation	Previous generic placement	Cu pla
Oreophryne flava	Parker, 1934	*Oreophryne*	*inc*
Oreophryne flavomaculata	Günther and Richards, 2016	*Oreophryne*	*inc*
Callula frontifasciata	Horst, 1883	*Oreophryne*	*inc*
Oreophryne furu	Günther, Richards, Tjaturadi, and Iskandar, 2009	*Oreophryne*	*inc*
Oreophryne geminus	Zweifel, Cogger, and Richards, 2005	*Oreophryne*	*inc*
Oreophryne habbemensis	Zweifel, Cogger, and Richards, 2005	*Oreophryne*	*inc*
Oreophryne hypsiops	Zweifel, Menzies, and Price, 2003	*Oreophryne*	*inc*
Oreophryne idenburgensis	Zweifel, 1956	*Oreophryne*	*inc*
Oreophryne jeffersoniana	Dunn, 1928	*Oreophryne*	*inc*
Oreophryne kampeni	Parker, 1934	*Oreophryne*	*inc*
Oreophryne kapisa	Günther, 2003	*Oreophryne*	*inc*
Sphenophryne mertoni	Roux, 1910	*Oreophryne*	*inc*

Original designation	Citation	Previous generic placement	Cu pla
Oreophryne minuta	Richards and Iskandar, 2000	*Oreophryne*	*inc*
Sphenophryne monticola	Boulenger, 1897	*Oreophryne*	*inc*
Oreophryne nicolasi	Richards and Günther, 2019	*Oreophryne*	*inc*
Oreophryne oviprotector	Günther, Richards, Bickford, and Johnston, 2012	*Oreophryne*	*inc*
Oreophryne pseudasplenicola	Günther, 2003	*Oreophryne*	*inc*
Oreophryne pseudunicolor	Günther and Richards, 2016	*Oreophryne*	*inc*
Oreophryne purari	Günther, Nagombi, and Richards, 2025	*Oreophryne*	*inc*
Oreophryne roedeli	Günther, 2015	*Oreophryne*	*inc*
Oreophryne rookmaakeri	Mertens, 1927	*Oreophryne*	*inc*
Oreophryne riyantoi	Putri, Trilaksono, Kurniati, Engilis, and Hamidy, 2023	*Oreophryne*	*inc*
Oreophryne sibilans	Günther, 2003	*Oreophryne*	*inc*
Oreophryne streiffeleri	Günther and Richards, 2012	*Oreophryne*	*inc*

Original designation	Citation	Previous generic placement	Cu pla
Oreophryne terrestris	Zweifel, Cogger, and Richards, 2005	*Oreophryne*	*inc*
Oreophryne unicolor	Günther, 2003	*Oreophryne*	*inc*
Oreophryne waira	Günther, 2003	*Oreophryne*	*inc*
Oreophryne wapoga	Günther, Richards, and Iskandar, 2001	*Oreophryne*	*inc*
Hylella wolterstorffi	Werner, 1901	*Oreophryne*	*inc*
Oreophryne zimmeri	Ahl, 1933	*Oreophryne*	*inc*

Conflict of Interest

The authors declare no conflict of interest.

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Data Availability Statement

The data underlying this article are available in the GenBank Nucleotide Database [accession numbers PV703761-PV703777] at https://www.ncbi.nlm.nih.gov/genbank/. The codes and datafiles needed to reproduce these results can be found at [GitHub repository] with DOI [supplied upon publication].

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